

PRACTICAL 8

Aim : Identify valid IPv6 addresses and if invalid IPv6 address then write reason for the same:

2001: db8: 3333: 4444: 5555: 6666: 7777: 8888
2001:0db8:0001:0000: 0000:0ab9:C0A8:0102
225.1.4.2
::
fe80:2030:31:24

Input :

Absolutely. Let's go through each address carefully, using the rules for IPv6.

IPv6 basics

An IPv6 address is 128 bits long and is normally written as 8 groups, called hextets, separated by colons (:).

Example :

2001:0db8:0000:0000:0000:ff00:0042:8329

Each hextet :

- Contains 1 to 4 hexadecimal digits.
- Hexadecimal digits can be :
 - 0–9
 - a–f
 - A–F
- There must normally be 8 hextets.
- :: can be used once to represent one or more groups of zeros.
- Spaces are not allowed inside an IPv6 address.

1. 2001: db8: 3333: 4444: 5555: 6666: 7777: 8888

Answer : Invalid

Let's examine it :

2001: db8: 3333: 4444: 5555: 6666: 7777: 8888

There are apparently 8 groups :

1. 2001
2. db8
3. 3333
4. 4444
5. 5555
6. 6666
7. 7777
8. 8888

The number of groups is correct, and all the characters are hexadecimal.

However, there are spaces after the colons.

For example :

2001: db8

^

The correct IPv6 notation cannot contain these spaces.

Correct form

Remove the spaces :

2001:db8:3333:4444:5555:6666:7777:8888

This is a valid IPv6 address.

Conclusion

Given address : Invalid

Reason : Spaces are present.

Without spaces : Valid

2. 2001:0db8:0001:0000:0000:0ab9:C0A8:0102

Answer : Valid

Let's divide it into its 8 groups :

2001 : 0db8 : 0001 : 0000 : 0000 : 0ab9 : C0A8 : 0102

Now check each group :

Group	Value	Valid?
1	2001	✓
2	0db8	✓
3	0001	✓
4	0000	✓
5	0000	✓
6	0ab9	✓
7	C0A8	✓
8	0102	✓

Every group contains 4 hexadecimal digits or fewer.

Notice :

C0A8 might look unusual because it contains letters, but that's completely fine.

In hexadecimal :

C = 12

0 = 0

A = 10

8 = 8

So C0A8 is a perfectly valid hexadecimal hextet.

There are also exactly 8 groups, so the address represents 128 bits.

Conclusion

2001:0db8:0001:0000:0000:0ab9:C0A8:0102 is a valid IPv6 address.

3. 225.1.4.2

Answer : Invalid as an IPv6 address

This address uses dots (.) :

225.1.4.2

IPv4 addresses are written using four decimal numbers separated by dots.

For example :

192.168.1.1

The given address :

225.1.4.2 follows the IPv4 format.

IPv6, on the other hand, uses hexadecimal groups separated by colons :

2001:db8::1

Therefore :

225.1.4.2 is not an IPv6 address.

Important distinction

It is actually a valid IPv4 address because:

- It has 4 decimal parts.
- Each part is between 0 and 255.
- The parts are separated by dots.

So:

As IPv4 : Valid

As IPv6 : Invalid

4. ::

Answer : Valid

This is an important special case.

IPv6 allows a special notation called zero compression.

The notation :

:: represents one or more groups of zeros.

For example:

0:0:0:0:0:0:0:0

can be shortened to ::

Therefore:

:: represents the IPv6 address :

0000:0000:0000:0000:0000:0000:0000:0000

This is called the unspecified IPv6 address.

Why is :: allowed?

Normally IPv6 requires 8 groups :

xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx

But :: can replace consecutive zero groups.

For example :

2001:0db8:0000:0000:0000:0000:0000:0001

can become : 2001:db8::1

Here, :: replaces five groups of zeros.

Important rule

:: can appear at most once in an IPv6 address.

For example :

2001::abcd::1234 is invalid because it contains two ::.

Conclusion

:: is a valid IPv6 address.

5. fe80:2030:31:24

Answer : Invalid

Let's count the groups :

fe80 : 2030 : 31 : 24

There are only 4 groups :

1. fe80
2. 2030
3. 31
4. 24

A normal IPv6 address requires 8 groups.

For example :

fe80:0000:0000:0000:0000:0000:0001

The given address has no :: to indicate that some zero groups have been omitted.

Could it be shortened?

Yes, but it would need ::.

For example, if the intended address were :

fe80:2030:31:24:0000:0000:0000:0000

it could be written as : fe80:2030:31:24::

That would be valid.

But :

fe80:2030:31:24 by itself does not contain enough information to represent all 128 bits.

Conclusion

Given address : Invalid

Reason : Only 4 hexets are present and there is no :: compression.

Final answer in exam format :

No.	Given Address	Valid / Invalid	Reason
1	2001: db8: 3333: 4444: 5555: 6666: 7777: 8888	Invalid	Spaces are present. IPv6 addresses cannot contain spaces.
2	2001:0db8:0001:0000:0000:0ab9:C0A8:0102	Valid	Contains exactly 8 valid hexadecimal hextets.
3	225.1.4.2	Invalid IPv6	It is an IPv4 address, using decimal numbers and dots.
4	::	Valid	:: is valid IPv6 zero compression and represents groups of zeros.
5	fe80:2030:31:24	Invalid	Only 4 hextets are present;